Concrete Software


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The number of concrete homes is growing, and now there is software available to correctly size HVAC equipment in these homes. In 1993, the above-grade concrete wall market for single-family detached homes represented 3% of new-home construction. Thanks to industry research and promotion efforts, the above-grade concrete wall market through 2003 was at 16%. Insulated concrete walls include insulating concrete form (ICF) walls, removable-form concrete walls, precast concrete walls, autoclaved aerated concrete (AAC) walls, and concrete masonry (CMU) walls. Houses constructed with concrete wall systems, if built properly, are quiet, comfortable, disaster resistant, sustainable, insect resistant, and energy efficient. Concrete's inherent thermal mass, the high level of insulation in some applications, and the low air infiltration of concrete walls all serve to increase energy efficiency.

A variety of studies have compared the energy performance of concrete homes to that of wood frame alternatives (see "ICFs Under the Microscope," *HE* Nov/Dec '02, p. 36 and "ICFs in North Texas," *HE* Nov/Dec '02, p. 39). One study sponsored by the Portland Cement Association (PCA) compared the energy use of identical contemporary houses with various exterior walls in 25 North American locations. Due to the thermal mass of concrete, houses with insulated concrete walls had lower heating and cooling costs than houses with wood frame walls. The consensus is that insulated concrete walls make it possible to downsize HVAC equipment as much as 40% compared to HVAC equipment in identical wood frame homes.

Unfortunately, widely used HVAC sizing methods, such as *Manual J* and *Manual S* and the ASHRAE *Handbook of Fundamentals* either are cumbersome to use or do not account for the thermal mass, high level of insulation, and low air infiltration of insulated concrete walls. Even worse, many builders and HVAC contractors size HVAC equipment based on a rule of thumb developed for wood frame homes that bases equipment size on floor area of living space. The net result is an inefficient HVAC system—one that is typically oversized. An oversized HVAC system will have a higher initial cost than a correctly sized system, and it will consume more energy than necessary to maintain thermostat setpoints. In addition, an oversized system will have a shortened on-time, which can lead to wide temperature swings and reduced thermal comfort. Air conditioning systems with short on-times do not remove enough moisture from the indoor environment. This can cause moisture problems and can cause or exacerbate respiratory illness in the occupants.

Based on the need for HVAC-sizing software specific to concrete homes, HUD sponsored an effort to compile available information regarding energy use in concrete homes and develop additional information as needed. It then used this information to develop a methodology to properly size HVAC equipment for concrete homes. The result of this effort—HVAC Sizing for Concrete Homes—is software intended for use by residential contractors to estimate the required heating-and-cooling system capacity for single-family concrete homes. The software is available for $59.95 from the Portland Cement Association and can be ordered through their Web site, www.cementhomes.com.

—John Gajda

John Gajda is a licensed professional engineer with Construction Technology Laboratories, Incorporated, in Skokie, Illinois. He managed several PCA-sponsored practical research projects on ICF walls, including studies on concrete consolidation, energy use, HVAC sizing, moisture susceptibility, and cold-weather construction.

For more information:

For more information regarding concrete homes, visit www.concretehomes.com.

For the PCA-sponsored study cited above, see Gajda, J. *Energy Use of Single-Family Houses with Various Exterior Walls*, PCA R&D Serial No. 2518 and PCA No. CD026. Skokie, Illinois: Portland Cement Association, 2001.